



# FROM GEOMETRY TO TOPOLOGY

Representing Buildings as Graphs

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# ABSTRACTION

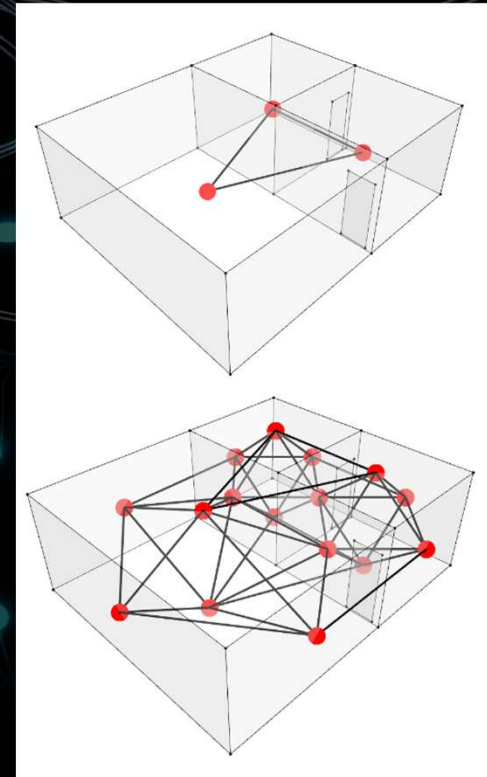
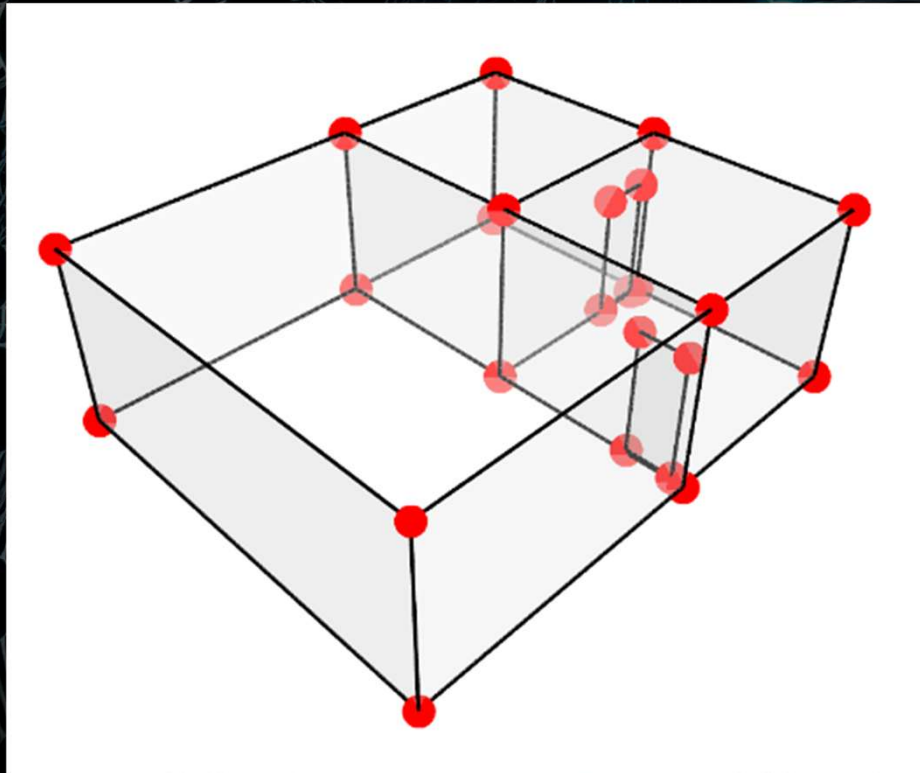
- Thinking of buildings as graphs is an act of abstraction.
- (From Wikipedia) “Abstraction is the process of generalizing rules and concepts from specific examples, literal (real or concrete) signifiers, first principles, or other methods. The result of the process, an abstraction, is a concept that acts as a common noun for all subordinate concepts and connects any related concepts as a group, field, or category.”
- Depending on context, intent, desired outcome, buildings can be converted into many types of graphs.
- Graphs can encapsulate one or multiple aspects of a building.



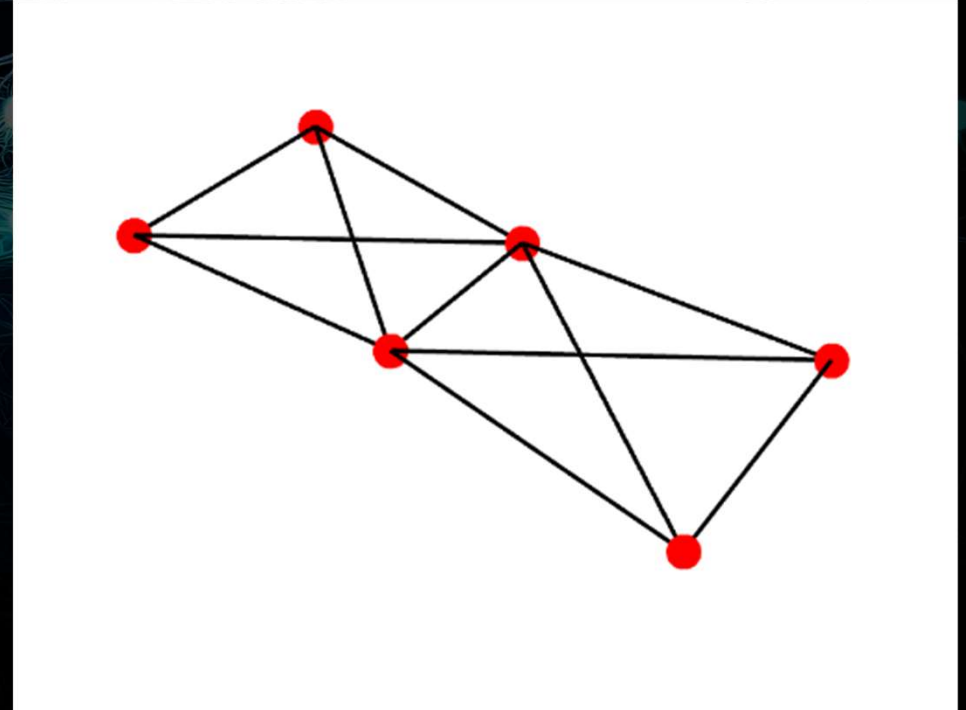
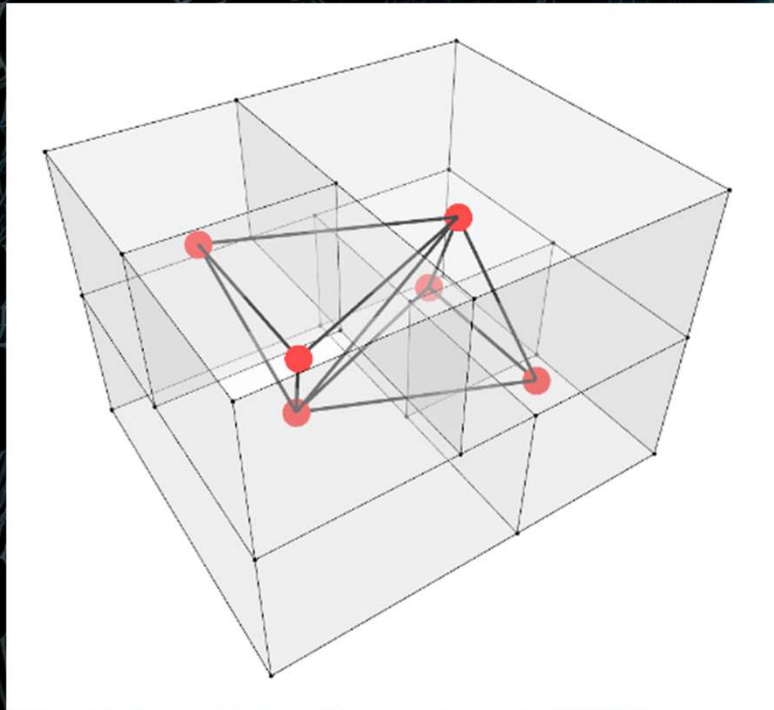
# GRAPH DIMENSIONS

- Primal vs Dual graphs
- Attributed vs non-attributed graphs
- Metric vs topological graphs
- Static vs dynamic graphs
- Deterministic vs probabilistic graphs
- Discrete vs continuous (sampled) graphs

# PRIMAL VS. DUAL GRAPHS



# METRIC VS. TOPOLOGICAL GRAPHS

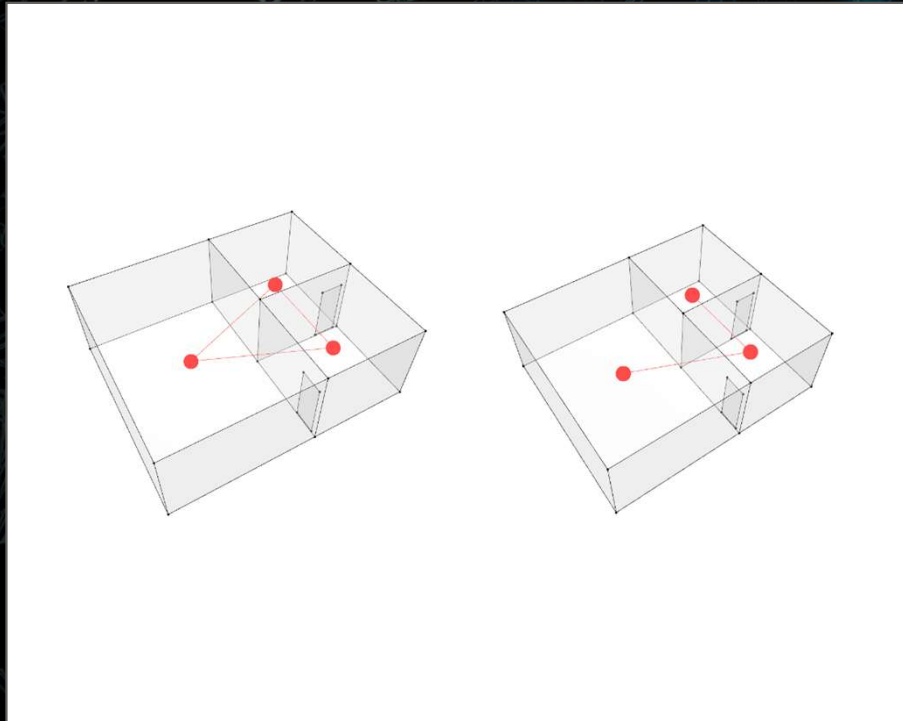




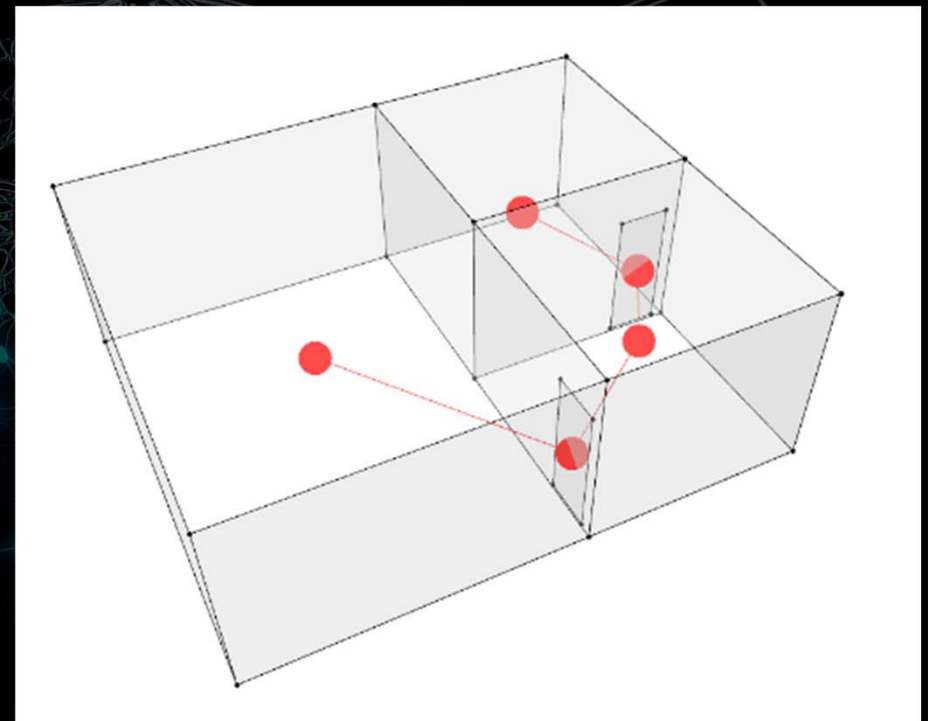
# SPATIAL / TOPOLOGICAL REPRESENTATIONS

- Room adjacency graph
  - Nodes: rooms/spaces
  - Edges: shared boundaries or direct access
- Access / circulation graph
  - Nodes: spaces or decision points
  - Edges: doors, corridors, paths of movement

## ADJACENCY GRAPH



## ACCESS/CIRCULATION GRAPH

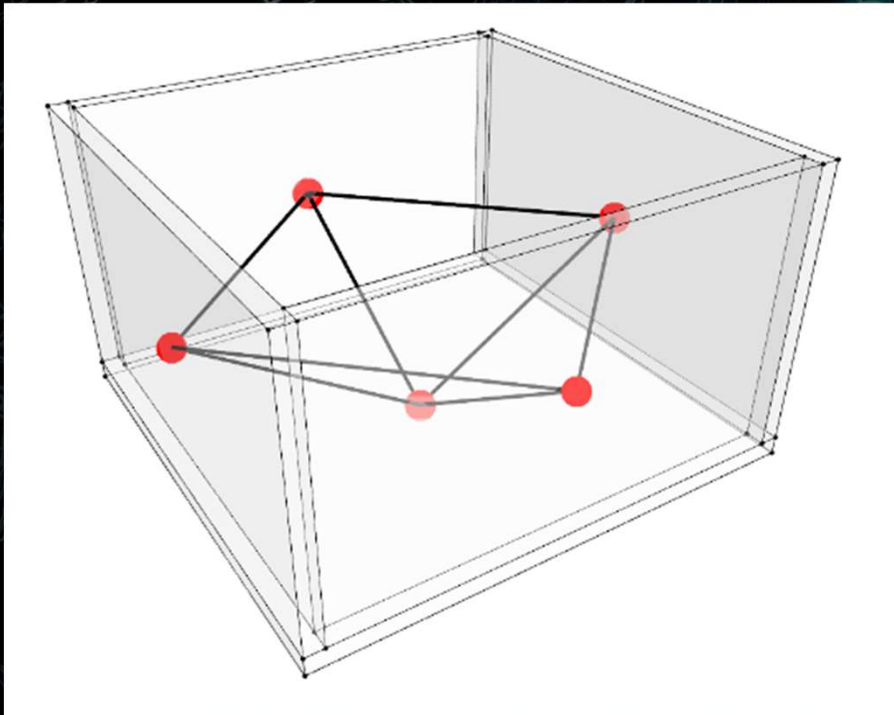


## GEOMETRIC / ELEMENT-BASED REPRESENTATIONS

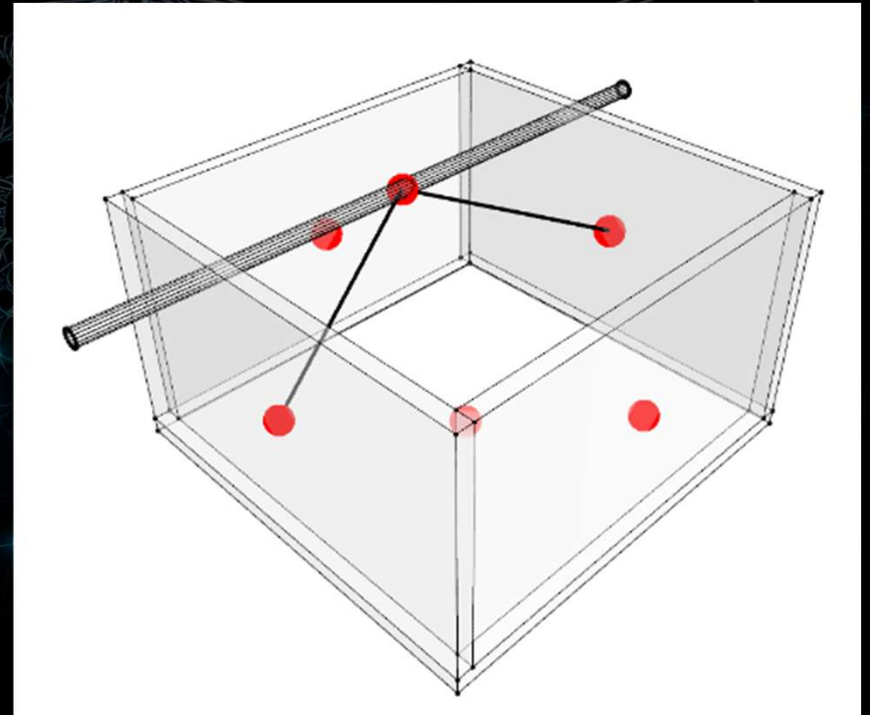
- Element connectivity graph
  - Nodes: building elements (walls, slabs, columns)
  - Edges: physical connections or adjacency



“TOUCHES”



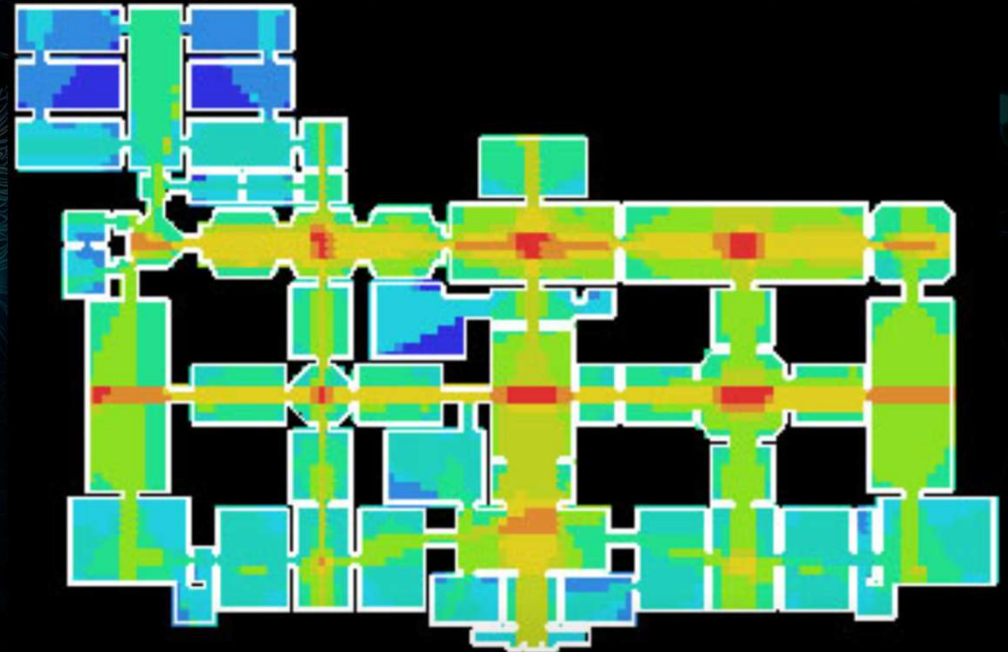
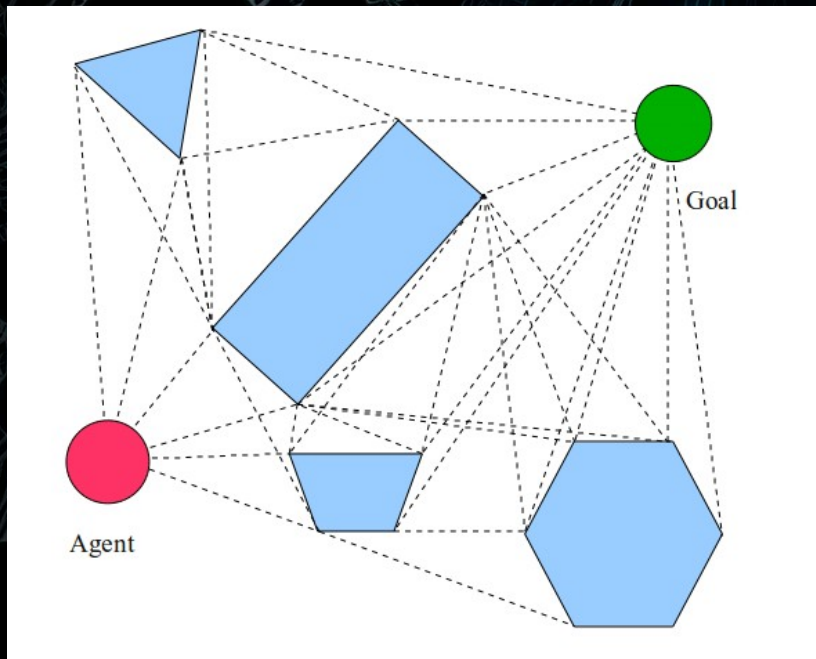
“OVERLAPS”



# SPATIAL / TOPOLOGICAL REPRESENTATIONS

- Visibility graph (VGA)
  - Nodes: sampled points in space
  - Edges: intervisibility (line-of-sight connections)

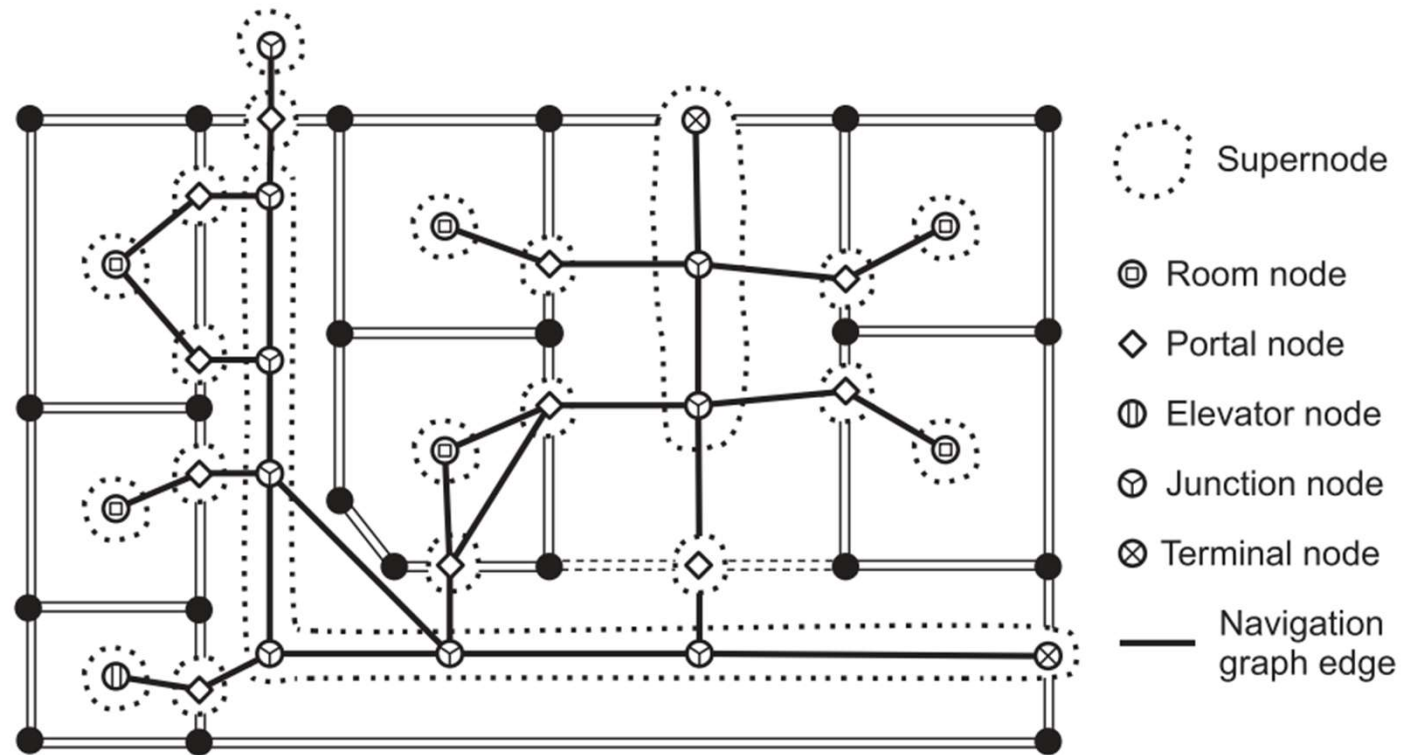
## VISIBILITY GRAPH





# CIRCULATION AND NAVIGATION GRAPHS

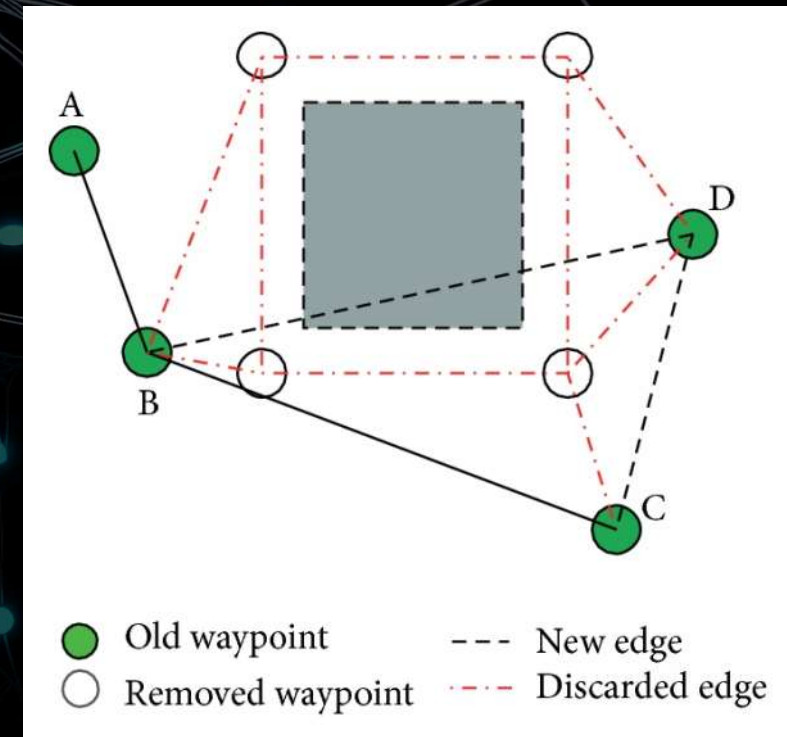
- Navigation graph
  - Nodes: walkable points or regions
  - Edges: traversable paths
- Waypoint graph
  - Nodes: key locations (doors, stairs, intersections)
  - Edges: feasible routes
- Evacuation graph
  - Nodes: spaces or exits
  - Edges: escape routes with capacities/weights



Supergraph generated from the skeleton navigation graph of

# WAYPOINT GRAPH

A waypoint graph represents navigable space using discrete points (waypoints) connected by edges that encode valid movement paths. Each node typically corresponds to a meaningful location in space (e.g., corridor intersections or room centroids), and edges represent traversable connections, often weighted by distance or cost. It is widely used in pathfinding and navigation because it reduces continuous space into a sparse, computationally efficient graph structure.





# EVACUATION DIAGRAM

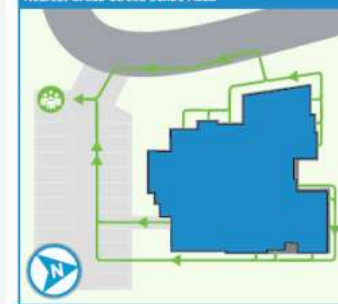


## CHILDCARE CENTRE - GROUND FLOOR

EVACUATION DIAGRAM NOT TO SCALE



**Childcare Centre**  
14 Childcare Way, Brisbane, QLD, 4000  
Nearest Cross-Street: Centre Road



### EVACUATION INSTRUCTIONS

- On Evacuation Signal, gather the children, emergency grab bags and medications.
- Educators to guide children to line up near exits and complete roll call and head count to ensure all children are accounted for.
- Once advised to Evacuate, guide the children to the nominated Assembly Area via the safest/quickest path outlined on evacuation diagrams.
- Lead Educator to complete final sweep and close doors to ensure no one is left behind. Alert the Chief Warden the room is all clear.
- Once at the Assembly Area, complete roll call and head count.
- Await further instructions from the Chief Warden.

Diagram Ref: AG - 104H - 7

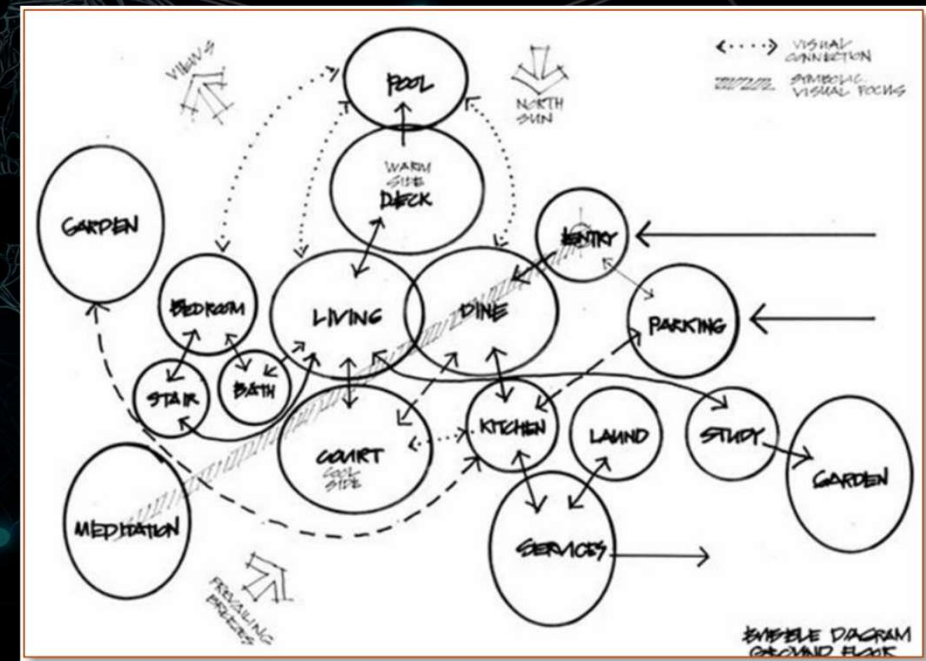
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Review Date: 02/2027

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# FUNCTIONAL / PROGRAMMATIC GRAPHS

- Program adjacency graph
  - Nodes: functions (kitchen, office, bedroom)
  - Edges: desired or actual adjacencies
- Zoning graph
  - Nodes: zones (public, private, service)
  - Edges: relationships or transitions
- Workflow graph
  - Nodes: activities
  - Edges: sequences or dependencies





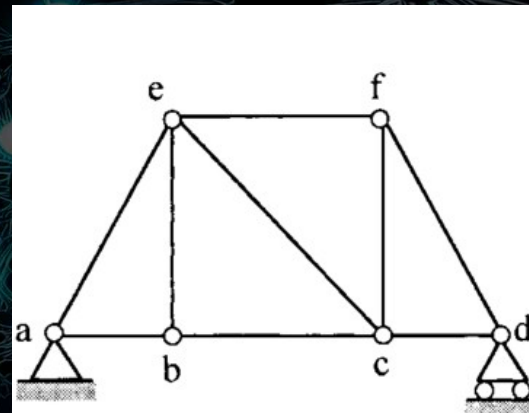
# SEMANTIC / INFORMATION-RICH GRAPHS

- BIM semantic graph
  - Nodes: building components with attributes
  - Edges: relationships (hosted by, connected to, belongs to)
- Knowledge graph of building systems
  - Nodes: entities (spaces, systems, users)
  - Edges: typed semantic relations
- GraphRAG-style architectural graph
  - Nodes: spaces/components with learned attributes
  - Edges: probabilistic or learned relationships

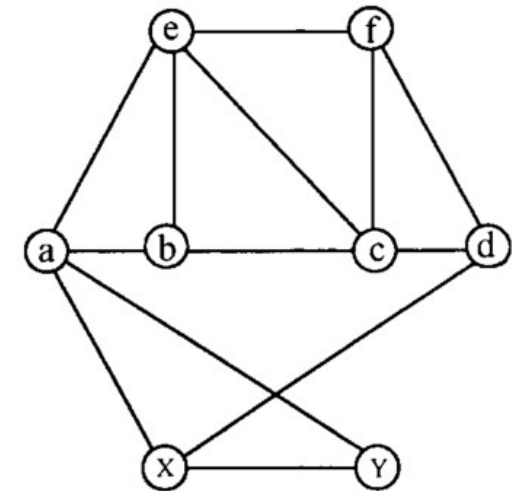


# PERFORMANCE / ANALYSIS GRAPHS

- Thermal network graph
  - Nodes: zones
  - Edges: heat transfer paths
- Structural graph
  - Nodes: joints
  - Edges: structural members
- Energy flow graph
  - Nodes: systems/components
  - Edges: energy exchange
- Acoustic propagation graph
  - Nodes: spaces
  - Edges: sound transmission paths



(a)



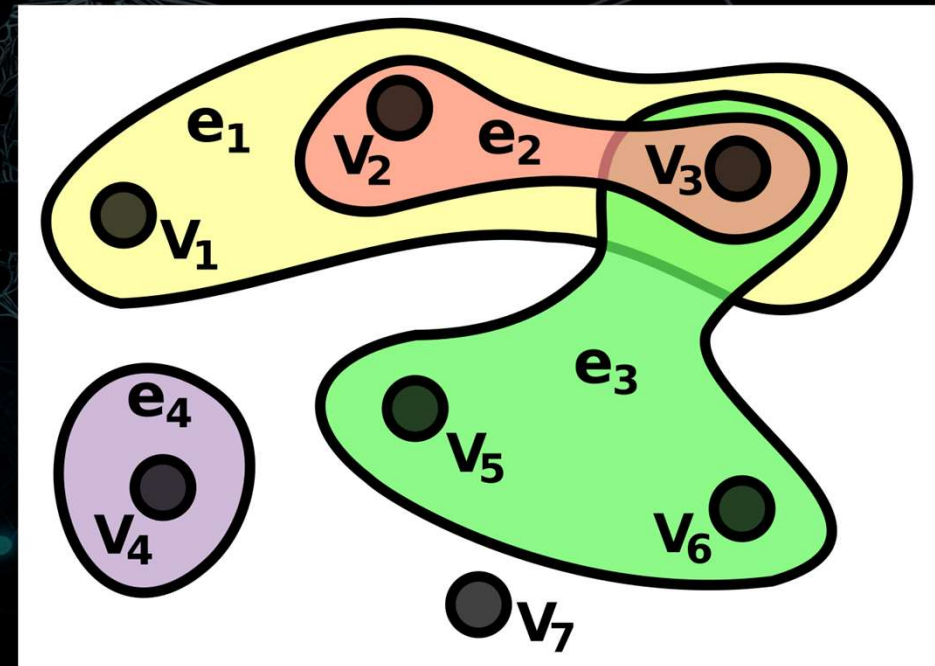
(b)

# HYPERGRAPHS

- A hypergraph is a generalisation of a graph in which relationships are not restricted to pairs.
- Nodes (vertices):
  - A set of discrete entities  $V=\{v_1, v_2, \dots, v_n\}$
- Hyperedges:
  - A set of subsets of nodes  $E=\{e_1, e_2, \dots, e_m\}$ , where each hyperedge
  - $e_i \subseteq V$  and can contain any number of nodes ( $\geq 1$ )



# HYPERGRAPHS





# THANK YOU

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